

# Special Senses

Vision · Hearing · Equilibrium · Olfaction · Gustation

---

- ▶ Overview of Special Senses
  - ▶ Vision — The Eye
  - ▶ Photoreception & Phototransduction
  - ▶ The Visual Pathway
  - ▶ Eye Movements & Ocular Motor Control
  - ▶ Hearing — The Ear
  - ▶ Auditory Transduction & Pathway
  - ▶ Disorders of Hearing
  - ▶ Vestibular System & Equilibrium
  - ▶ Olfaction (Smell)
  - ▶ Gustation (Taste)
- 

**everythingabuad.com**

PHS 220 — Physiology

## § TABLE OF CONTENTS

1. Overview of Special Senses
  - Classification
  - Cranial nerve summary
2. Vision — The Eye
  - Anatomy
  - Optical system
  - Refractive errors
  - Intraocular pressure
3. Photoreception & Phototransduction
  - Rods and cones
  - Visual pigments
  - Dark/light adaptation
4. The Visual Pathway
  - Retina to cortex
  - Visual field defects
  - Pupillary reflexes
5. Eye Movements & Ocular Motor Control
  - Conjugate gaze
  - Vergence
  - Saccades and smooth pursuit
6. Hearing — The Ear
  - External ear
  - Middle ear
  - Inner ear (cochlea)
7. Auditory Transduction & Pathway
  - Hair cells
  - Coding of frequency
  - Auditory cortex
8. Disorders of Hearing
  - Sensorineural
  - Conductive
  - Tests
9. Vestibular System & Equilibrium
  - Semicircular canals
  - Otolith organs
  - Pathway
  - Nystagmus
10. Olfaction (Smell)
  - Receptor neurons

- Olfactory pathway
- Clinical notes

## 11. Gustation (Taste)

- Taste buds
  - Five primary tastes
  - Pathway
-

## §1 OVERVIEW OF SPECIAL SENSES

The special senses are highly specialised sensory systems that provide detailed information about the external environment. Unlike general somatic senses (touch, pain, proprioception), the special senses are confined to specific receptor organs in the head and are carried by dedicated cranial nerves to specialised cortical areas.

| Sense               | Receptor                                                                       | Cranial Nerve                                     | Primary Cortex                                                           | Key Feature                                                                                                                         |
|---------------------|--------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Vision              | Photoreceptors (rods & cones) in retina                                        | CN II (Optic)                                     | Primary visual cortex BA17 (calcarine sulcus, occipital)                 | Only sense with a relay nucleus in thalamus (LGN) — but also direct connections (pupil reflex, circadian)                           |
| Hearing             | Hair cells in organ of Corti (cochlea)                                         | CN VIII (Vestibulocochlear — cochlear division)   | Primary auditory cortex BA41/42 (Heschl's gyri, superior temporal)       | Tonotopic organisation from cochlea to cortex; bilateral representation protects from unilateral lesion                             |
| Equilibrium/Balance | Hair cells in semicircular canals (angular) and otolith organs (linear)        | CN VIII (Vestibulocochlear — vestibular division) | Parietal-insular vestibular cortex (PIVC)                                | Crossed and uncrossed projections; strong connections to cerebellum and eye movement centres                                        |
| Olfaction           | Olfactory receptor neurons (CN I — unique: bipolar; replaced every 30–60 days) | CN I (Olfactory)                                  | Olfactory cortex (piriform, entorhinal) — only sense bypassing thalamus! | Direct projections to limbic system (amygdala, entorhinal → hippocampus) — explains powerful emotional/memory associations of smell |
| Taste (Gustation)   | Taste receptor cells in taste buds on tongue/palate/epiglottis                 | CN VII, IX, X                                     | Primary gustatory cortex (insula / frontal operculum)                    | 5 primary tastes: sweet, sour, salty, bitter, umami (savory)                                                                        |

## §2 VISION — THE EYE

### 2.1 Anatomy of the Eyeball

| Layer/Structure          | Description                                                                                                | Function                                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Outer Fibrous Coat       |                                                                                                            |                                                                                                                               |
| — Cornea (anterior 1/6)  | Transparent, avascular, 5 layers (epithelium, Bowman's, stroma, Descemet's, endothelium); regularly curved | Provides ~70% of total refractive power (~43D); light enters here; avascular (receives O <sub>2</sub> from tears and aqueous) |
| — Sclera (posterior 5/6) | White opaque fibrous tissue; merges with cornea at limbus                                                  | Protective outer shell; attachment point for extraocular muscles                                                              |
| Middle Uveal Layer       |                                                                                                            |                                                                                                                               |

| Layer/Structure      | Description                                                                                                                                                   | Function                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| — Choroid            | Highly vascular; contains melanin (anti-reflection)                                                                                                           | Blood supply to outer retina (photoreceptors); heat dissipation                                                                |
| — Ciliary Body       | Ring of smooth muscle + ciliary processes; attached to lens via zonule fibres                                                                                 | Ciliary muscle contracts → zonule fibres relax → lens rounds → ACCOMMODATION; produces aqueous humour (from ciliary processes) |
| — Iris               | Coloured ring with central pupil aperture; dilator (radial, sympathetic $\alpha_1$ ) + constrictor (circular, PSNS M3)                                        | Controls light entering eye; pupil dilation/constriction                                                                       |
| Inner Retina         |                                                                                                                                                               |                                                                                                                                |
| — Retina (10 layers) | Inner neural layer; extends from ora serrata to optic disc                                                                                                    | Phototransduction; initial visual processing; ganglion cell axons form optic nerve                                             |
| — Macula             | Central ~5mm retina; highest cone density; avascular centre (fovea)                                                                                           | Highest acuity vision; central colour vision; the fovea is the point of best vision                                            |
| — Fovea Centralis    | Central pit of macula; ~0.35 mm; almost exclusively cones; other retinal layers displaced                                                                     | ~0.35 mm diameter; greatest visual acuity; one-to-one cone:ganglion cell ratio                                                 |
| — Optic Disc         | Nasal retina; no photoreceptors ("blind spot"); optic nerve exits; central retinal artery/vein                                                                | No vision here (physiological blind spot at 15° nasal to fixation)                                                             |
| Optical Media        |                                                                                                                                                               |                                                                                                                                |
| — Aqueous Humour     | Clear fluid in anterior/posterior chambers; produced by ciliary body (active secretion); drains via trabecular meshwork → Canal of Schlemm → episcleral veins | Maintains IOP (10-21 mmHg); nourishes avascular cornea and lens; obstruction → glaucoma                                        |
| — Lens               | Transparent, biconvex, avascular; nucleus (older fibres) + cortex; enclosed in capsule; no pain fibres                                                        | Variable refraction via accommodation (ciliary muscle); provides ~20-25D adjustable; opacification = cataract                  |
| — Vitreous Humour    | Gel (98% water + collagen + hyaluronic acid); fills posterior segment; not regenerated                                                                        | Maintains globe shape; supports retina against choroid; floaters = condensations                                               |

## 2.2 Optics — Refraction & Accommodation

The eye is a compound optical system with a total refractive power of approximately +60 dioptres (D) when focused at infinity. The cornea contributes ~43D (fixed) and the lens provides ~20D (variable — adjusted by accommodation). Light is refracted to converge precisely on the fovea.

| Refractive Error                       | Optical Problem                                              | Point of Focus                              | Vision Defect                                                             | Correction                                            |
|----------------------------------------|--------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------|
| Emmetropia (normal)                    | None — parallel rays focus exactly on retina                 | On retina (fovea)                           | Normal                                                                    | None needed                                           |
| Myopia (short-sight)                   | Eyeball too long OR cornea/lens too curved                   | Anterior to retina                          | Blurred distance vision; clear near vision                                | Concave (diverging) lens — negative dioptre           |
| Hypermetropia / Hyperopia (long-sight) | Eyeball too short OR cornea/lens too flat                    | Posterior to retina (would converge behind) | Blurred near vision; distance variable; eye uses accommodation to correct | Convex (converging) lens — positive dioptre           |
| Astigmatism                            | Cornea or lens with unequal curvature in different meridians | Multiple focal points                       | Blurred and distorted vision at all distances                             | Cylindrical lens; toric contact lens; LASIK           |
| Presbyopia                             | Age-related loss of lens elasticity (lens hardens with age)  | Cannot accommodate near objects             | Near vision blur; distance preserved; onset ~45 years                     | Reading glasses (convex); bifocals; multifocal lenses |

#### ACCOMMODATION TRIAD

**When focusing on a near object, three simultaneous responses occur (controlled by CN III and the Edinger-Westphal parasympathetic nucleus):**

- 1. Accommodation: Ciliary muscle contracts → zonule fibres slacken → lens rounds (↑ curvature → ↑ refractive power → near focus).**
- 2. Pupillary constriction (Miosis): Circular pupillary constrictor (M3) contracts → smaller pupil → ↑ depth of field → reduces spherical aberration.**
- 3. Convergence: Both eyes turn medially (medial rectus CN III) to maintain binocular single image of the near object.**

## 2.3 Intraocular Pressure & Glaucoma

Normal intraocular pressure (IOP) is 10–21 mmHg, maintained by the balance between aqueous humour production (by ciliary processes) and drainage (via trabecular meshwork → Schlemm's canal → episcleral veins). In **open-angle glaucoma** (commonest type), resistance in the trabecular meshwork increases → IOP rises → optic nerve compression → progressive retinal ganglion cell loss → visual field loss (arcuate scotomata, then peripheral loss, sparing central vision until late).

## §3 PHOTORECEPTION & PHOTOTRANSDUCTION

### 3.1 Rods vs Cones

| Property              | Rods                                                                             | Cones                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Number                | ~120 million per retina                                                          | ~6-7 million per retina                                                                                                                  |
| Distribution          | Peripheral retina; absent from fovea; highest density at ~20° eccentricity       | Fovea (highest density); sparse in periphery                                                                                             |
| Sensitivity           | Extremely sensitive (single photon can signal); function in dim light (scotopic) | Less sensitive; function in bright light (photopic)                                                                                      |
| Colour discrimination | None (one pigment — rhodopsin); achromatic                                       | Yes — 3 cone types (S=blue, M=green, L=red); trichromacy                                                                                 |
| Visual acuity         | Low (many rods → one ganglion cell — convergence)                                | High (1:1 ratio of foveal cones to ganglion cells at fovea)                                                                              |
| Photopigment          | Rhodopsin ( $\lambda_{\max}$ = 498 nm)                                           | Photopsin I (red, $\lambda_{\max}$ 560 nm); Photopsin II (green, $\lambda_{\max}$ 530 nm); Photopsin III (blue, $\lambda_{\max}$ 420 nm) |
| Adaptation            | Dark-adapt fully in 20-30 min; light-adapt in seconds                            | Light-adapt rapidly; dark-adapt in 5-7 min                                                                                               |

### 3.2 Phototransduction Cascade (in Rods)

- **Dark state:** cGMP is high → cGMP-gated cation channels open →  $\text{Na}^+$  and  $\text{Ca}^{2+}$  flow in → photoreceptor is DEPOLARISED (-40 mV) → continuous glutamate release onto bipolar cells (inhibits ON bipolar cells; excites OFF bipolar cells).
- **Photon absorption:** A photon is absorbed by rhodopsin (opsin + 11-cis retinal, a vitamin A aldehyde) → 11-cis retinal isomerises to all-trans retinal → conformational change in opsin → activated rhodopsin ( $\text{R}^*$ ).
- **Transducin activation:**  $\text{R}^*$  activates transducin (Gt — a G-protein) →  $\text{Gt}\alpha$  subunit activated → activates phosphodiesterase (PDE).
- **cGMP hydrolysis:** PDE hydrolyses cGMP → cGMP falls → cGMP-gated channels CLOSE → less  $\text{Na}^+$  and  $\text{Ca}^{2+}$  entering → rod HYPERPOLARISES (toward -70 mV).
- **Reduced glutamate release:** Hyperpolarisation → less  $\text{Ca}^{2+}$  influx at terminal → less glutamate released → signals transmitted to bipolar then ganglion cells (ON pathway activated).
- **Recovery:** Rhodopsin kinase phosphorylates  $\text{R}^*$  → arrestin binds → deactivates  $\text{R}^*$ ;  $\text{Gt}\alpha$ -GTPase hydrolyses GTP; PDE deactivates; guanylyl cyclase (GC, stimulated by falling  $\text{Ca}^{2+}$  via GCAP) resynthesises cGMP → recovery to dark state.

#### COLOUR VISION & COLOUR BLINDNESS

Normal trichromatic vision uses three cone types. Colour is perceived from the ratio of activation of L, M, S cones (opponent processing: R-G, B-Y, Light-Dark channels).

**Red-green colour blindness** (~8% males, 0.5% females — X-linked): Most common type; loss or shift in L (protan) or M (deutan) cone pigment absorption spectrum. Males are hemizygous for X-linked opsin genes — hence higher prevalence.

**Achromatopsia:** Complete absence of cone function; requires bright light for poor vision only; completely colour blind.

## §4 THE VISUAL PATHWAY

| Station                                                                | Anatomy                                                                                  | Key Points                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Photoreceptors<br>→ Bipolar cells →<br>Retinal Ganglion<br>Cells (RGC) | Vertical pathway in retina;<br>lateral modulation by<br>horizontal and amacrine<br>cells | RGC axons form optic nerve (CN II); ON-centre / OFF-centre RGC types; M (magno — motion/depth) and P (parvo — colour/form) channels begin here                                                                                             |
| Optic Nerve (CN II)                                                    | RGC axons exit at optic disc<br>→ optic nerve → optic<br>chiasm                          | Nasal fibres decussate at chiasm; temporal fibres stay ipsilateral; ~1.2 million RGC axons per eye                                                                                                                                         |
| Optic Chiasm                                                           | X-shaped crossing at base<br>of hypothalamus; anterior<br>to pituitary stalk             | Nasal retina fibres (representing temporal visual field) cross → contralateral optic tract. Temporal fibres do NOT cross. Result: each optic tract carries the contralateral visual field from both eyes                                   |
| Optic Tract                                                            | From chiasm to lateral<br>geniculate nucleus (LGN)                                       | Left optic tract = left visual cortex = right visual field from both eyes                                                                                                                                                                  |
| Lateral<br>Geniculate<br>Nucleus (LGN)                                 | In dorsal thalamus (6<br>layers); retinotopic<br>organisation                            | Synapse here; layers 1,2 = M pathway (motion); layers 3-6 = P pathway (colour/form); koniocellular layers (K) between them. Most RGC axons synapse here.                                                                                   |
| Optic Radiation                                                        | From LGN → primary visual<br>cortex; via internal capsule                                | Superior fibres → superior bank of calcarine (lower field); Inferior fibres (Meyer's loop, temporal lobe) → inferior bank (upper field). Meyer's loop lesion (temporal lobectomy) → contralateral upper quadrantanopia ("pie in the sky"). |
| Primary Visual<br>Cortex (V1,<br>BA17)                                 | Banks of calcarine sulcus,<br>occipital lobe                                             | Striate cortex; retinotopic; upper bank = lower visual field; lower bank = upper visual field; large macular representation (macular sparing in occipital stroke — dual blood supply from MCA + PCA)                                       |

### 4.1 Visual Field Defects — Anatomical Localisation

| Lesion Site                         | Visual Field Defect                                | Typical Cause                                                         |
|-------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------|
| Optic nerve (pre-chiasm)            | Unilateral blindness (ipsilateral eye only)        | Optic neuritis (MS); AION; orbital tumour                             |
| Optic chiasm (central)              | Bitemporal hemianopia (tunnel vision)              | Pituitary adenoma; craniopharyngioma (compress crossing nasal fibres) |
| Optic chiasm (lateral/perichiasmal) | Binasal hemianopia (rare)                          | Bilateral carotid aneurysms; meningioma pressing on lateral chiasm    |
| Optic tract (post-chiasm)           | Contralateral homonymous hemianopia (incongruous)  | Less common — trauma, tumour                                          |
| Meyer's loop (temporal lobe)        | Contralateral superior quadrantanopia (homonymous) | Temporal lobectomy (e.g., for epilepsy); temporal lobe tumour         |



| Lesion Site              | Visual Field Defect                                      | Typical Cause                                                           |
|--------------------------|----------------------------------------------------------|-------------------------------------------------------------------------|
| Parietal optic radiation | Contralateral inferior quadrantanopia (homonymous)       | Parietal tumour/infarction                                              |
| Complete optic radiation | Contralateral homonymous hemianopia (congruous)          | MCA territory infarction                                                |
| Occipital cortex (V1)    | Contralateral homonymous hemianopia with MACULAR SPARING | PCA territory infarction (macular area often spared by MCA collaterals) |

## §5 EYE MOVEMENTS & OCULAR MOTOR CONTROL

| Movement Type                 | Purpose                                                                                                    | Control Centre                                        | Cranial Nerves                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------|
| Saccades                      | Rapid, voluntary eye movements to direct gaze to a new target (ballistic — cannot be corrected mid-flight) | Frontal Eye Fields (FEF, BA8); Superior Colliculus    | CN III, IV, VI via brainstem PPRF (horizontal) and riMLF (vertical) |
| Smooth Pursuit                | Track a slowly moving target; requires moving target                                                       | Middle Temporal area (V5/MT); parieto-occipital areas | CN III, IV, VI                                                      |
| Vergence                      | Convergence (near) or divergence (far) — eyes move in opposite directions for binocular fusion             | Convergence centre in midbrain                        | CN III (medial rectus bilateral)                                    |
| Vestibulo-Ocular Reflex (VOR) | Stabilises gaze during head movement; compensatory eye movement opposite to head rotation                  | Vestibular nuclei → MLF → CN III/IV/VI nuclei         | CN III, IV, VI; CN VIII (afferent)                                  |
| Optokinetic Reflex (OKR)      | Stabilises full-field visual scene during body movement; slow phase follows, fast phase resets             | Occipital cortex → brainstem                          | CN III, IV, VI                                                      |

### INTERNUCLEAR OPHTHALMOPLEGIA (INO)

The **Medial Longitudinal Fasciculus (MLF)** connects the CN VI nucleus (horizontal gaze centre) to the contralateral CN III nucleus (medial rectus), coordinating conjugate horizontal gaze. **INO** results from MLF lesion: on attempted lateral gaze, the ipsilateral eye fails to adduct (medial rectus palsy), while the contralateral eye abducts with nystagmus. **Bilateral INO in a young person = MS until proven otherwise.** Unilateral INO — consider brainstem stroke or tumour.

## §6 HEARING — THE EAR

### 6.1 Structure & Function: External, Middle, Inner Ear

| Part                          | Structures                                                                                                                                                                       | Function                                                                                                                                                                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| External Ear                  | Pinna (auricle) + External Auditory Meatus (EAM, ~2.5 cm) + Tympanic Membrane (TM)                                                                                               | Collects and channels sound waves to TM; pinna assists sound localisation; EAM resonates ~3.5 kHz; TM vibrates and transmits to ossicles                                                                                                |
| Middle Ear (Tympanic Cavity)  | Three ossicles: Malleus (handle on TM) → Incus → Stapes (oval window); Eustachian tube; tensor tympani (CN V3) + stapedius (CN VII)                                              | Impedance matching between air (low impedance) and perilymph (high impedance); 22× mechanical advantage (area ratio + lever) prevents ~99.9% energy loss at air-fluid interface; acoustic reflex (stapedius reflex dampens loud sounds) |
| Inner Ear — Cochlea           | Coiled ~2.5 turns; 3 fluid-filled chambers: Scala vestibuli (perilymph) + Scala media/cochlear duct (endolymph) + Scala tympani (perilymph); oval window (stapes) + round window | Transduction of sound vibration to electrical signals; frequency analysis (tonotopic — high freq at base, low freq at apex); hair cells on Basilar Membrane (organ of Corti)                                                            |
| Inner Ear — Vestibular System | 3 semicircular canals (SSC: anterior, posterior, lateral) + Utricle + Saccule; filled with endolymph; hair cells in cristae (SSC) and maculae (utricle/sacculle)                 | Detect angular acceleration (SSC) and linear acceleration/gravity (utricle/sacculle)                                                                                                                                                    |

### 6.2 Cochlear Mechanics & Frequency Analysis

Sound waves transmitted via ossicles cause the stapes footplate to piston in and out of the oval window, creating pressure waves in the perilymph. These waves travel along the basilar membrane (BM), which has graded mechanical properties along its length:

| BM Region               | Mechanical Properties | Best Frequency (BF)         | Sound Pitch                           |
|-------------------------|-----------------------|-----------------------------|---------------------------------------|
| Base (near oval window) | Narrow, thick, stiff  | High frequency (~20,000 Hz) | High-pitched sounds (e.g., birdsong)  |
| Apex (near helicotrema) | Wide, thin, flexible  | Low frequency (~20 Hz)      | Low-pitched sounds (e.g., bass notes) |

Each location on the BM resonates maximally at its characteristic frequency (travelling wave theory — von Békésy, Nobel 1961). The outer hair cells (OHC) actively amplify (electromotility via prestin protein) the travelling wave at its peak location, sharpening frequency selectivity by ~100×.

## §7 AUDITORY TRANSDUCTION & PATHWAY

## 7.1 Hair Cell Mechanoelectric Transduction

| Step                                    | Event                                                                                                                                                                                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. BM deflection                        | Travelling wave causes BM displacement at peak frequency location                                                                                                                                                      |
| 2. Shearing motion                      | BM movement causes shearing between TM (tectorial membrane) and reticular lamina → stereocilia of inner hair cells (IHC) deflected toward tallest stereocilium (kinocilium in vestibular; toward modiolus in auditory) |
| 3. Tip links stretch                    | Extracellular tip links connect adjacent stereocilia → deflection TOWARD tall row → tip links pulled → MET (mechanoelectric transducer) channels OPEN                                                                  |
| 4. K <sup>+</sup> influx                | Endolymph rich in K <sup>+</sup> (+80 mV endocochlear potential maintained by stria vascularis); K <sup>+</sup> flows IN through MET channels → depolarisation of IHC (~-45 mV)                                        |
| 5. Ca <sup>2+</sup> influx & NT release | Depolarisation → voltage-gated Ca <sup>2+</sup> channels at basolateral surface open → Ca <sup>2+</sup> in → exocytosis of glutamate onto afferent dendrites of spiral ganglion neurons (type I, innervate IHC only)   |
| 6. AP in CN VIII                        | Glutamate → AMPA receptors on spiral ganglion dendrites → AP → cochlear division CN VIII → cochlear nuclei                                                                                                             |

## 7.2 The Auditory Pathway

| Station                         | Location                                           | Processing Function                                                                                                                                                                  |
|---------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spiral Ganglion (1st-order)     | Within modiolus of cochlea                         | IHC (95%) → afferent type I neurons; OHC (5%) → type II (function unclear)                                                                                                           |
| Cochlear Nuclei (2nd-order)     | Pontomedullary junction; dorsal and ventral        | First binaural interaction begins; tonotopic maintained; first central processing of timing and spectral features                                                                    |
| Superior Olivary Complex (SOC)  | Pons (first major decussation of auditory pathway) | Binaural processing: interaural time difference (ITD) in medial superior olive (MSO) → sound localisation azimuth; interaural level difference (ILD) in lateral superior olive (LSO) |
| Inferior Colliculus (IC)        | Midbrain tectum                                    | Integration of binaural cues; gating of auditory input; descending control                                                                                                           |
| Medial Geniculate Nucleus (MGN) | Thalamus (auditory relay)                          | Tonotopic relay to cortex; gating of attention; 3 divisions (ventral — specific; dorsal — non-specific; medial — multimodal)                                                         |
| Primary Auditory Cortex (A1)    | BA41/42; Heschl's gyri; superior temporal plane    | Conscious hearing; tonotopic map; sound identification; speech processing; bilateral representation means unilateral cortical lesions rarely cause deafness                          |

## §8 DISORDERS OF HEARING

| Type                              | Definition                                                                                       | Causes                                                                                                                                                                                 | Rinne Test (512 Hz)                                                     | Weber Test (512 Hz)                                                                | Treatment                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Conductive Hearing Loss (CHL)     | Loss due to defect in transmission of sound from environment to cochlea (external or middle ear) | Wax impaction; perforated TM; otitis media (glue ear); otosclerosis (stapedial fixation); ossicular disruption; cholesteatoma                                                          | BC > AC (negative Rinne) — bone conduction bypasses middle ear problem  | Lateralises to AFFECTED ear (vibration heard better where conduction is "boosted") | Remove wax; grommets (glue ear); surgery (tympanoplasty, stapedectomy/plasty); hearing aid |
| Sensorineural Hearing Loss (SNHL) | Loss due to cochlear hair cell damage or retrocochlear (CN VIII/auditory cortex) pathology       | Noise exposure (most common); Presbycusis (age); ototoxic drugs (gentamicin, cisplatin, furosemide); Ménière's disease; acoustic neuroma (CN VIII); congenital (CMV, rubella, genetic) | AC > BC (positive Rinne) — but both reduced (BC not better than normal) | Lateralises to UNAFFECTED ear (cochlea problem → affected side processed less)     | Hearing aid; cochlear implant (severe/profound); treat underlying cause                    |
| Mixed Hearing Loss                | Combined CHL + SNHL                                                                              | Chronic otitis media; trauma                                                                                                                                                           | BC > AC but both reduced                                                | Variable                                                                           | Combination approach                                                                       |

#### AUDIOLOGICAL TESTS

**Pure Tone Audiogram (PTA):** Plots hearing threshold at frequencies 250–8000 Hz for air conduction (AC) and bone conduction (BC). AC-BC gap >10–15 dB = conductive component. Air conduction tests whole pathway; bone conduction bypasses outer and middle ear → tests inner ear and CN VIII only.

**Tympanometry:** Measures TM compliance at different pressure — flat type B tympanogram = middle ear effusion (glue ear); negative pressure = Eustachian tube dysfunction.

**Otoacoustic Emissions (OAEs):** Sounds generated by active OHC electromotility; present in normal hearing; absent in OHC damage → used in neonatal hearing screening.

## §9 VESTIBULAR SYSTEM & EQUILIBRIUM

| Structure                     | Sensor Type                               | Detects                                                                                                         | Stimulus                                          | Mechanism                                                                                                                                                                                                             |
|-------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semicircular Canals (3 pairs) | Angular velocity detector                 | Angular (rotational) acceleration in three planes: anterior (pitch), posterior (roll), lateral/horizontal (yaw) | Rotation of head                                  | Endolymph in canal lags behind head rotation → cupula (gelatinous barrier over crista) deflects → bends hair cells in ampulla → afferent firing changes (increased or decreased depending on direction of deflection) |
| Utricle                       | Linear acceleration detector (horizontal) | Horizontal linear acceleration; head tilt                                                                       | Gravity / linear acceleration in horizontal plane | Otoconia (calcium carbonate crystals on otolithic membrane) slide over macula → shear hair cells; polarised toward striola                                                                                            |

| Structure | Sensor Type                             | Detects                                             | Stimulus                                   | Mechanism                                          |
|-----------|-----------------------------------------|-----------------------------------------------------|--------------------------------------------|----------------------------------------------------|
| Sacculle  | Linear acceleration detector (vertical) | Vertical linear acceleration (gravity when upright) | Gravity in vertical plane; bouncing motion | Same otolith mechanism; macula oriented vertically |

## 9.1 The Vestibular Pathway

| Station                                          | Connections & Functions                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vestibular nerve (CN VIII — vestibular division) | Bipolar neurons in Scarpa's ganglion; primary afferents → brainstem vestibular nuclei (superior, medial, lateral, inferior) and cerebellum (vestibulocerebellum)                                                                                                                                                           |
| Vestibular Nuclei                                | 1) → Cerebellum (vestibulocerebellum/flocculonodular lobe) — coordination; 2) → Spinal cord (vestibulospinal tract) — postural reflexes, antigravity tone; 3) → CN III/IV/VI via MLF (Vestibulo-Ocular Reflex — VOR); 4) → Thalamus → Cortex (conscious orientation); 5) → Autonomic (nausea, vomiting in motion sickness) |
| Vestibulo-Ocular Reflex (VOR)                    | Head rotation detected → compensatory eye movement in opposite direction → gaze stabilisation during head movement. Gain normally 1.0; tested by head impulse test (HIT). Abnormal VOR → catch-up saccade (positive HIT) = peripheral vestibular deficit                                                                   |

### NYSTAGMUS — PERIPHERAL vs CENTRAL

**Peripheral (labyrinthine/CN VIII):** Horizontal/torsional nystagmus; fatigable (habituates); unidirectional; fast phase beats AWAY from lesion; suppressed by visual fixation; associated with vertigo, nausea. Causes: BPPV (most common; otoconia migration to posterior SSC); vestibular neuritis; Ménière's disease.

**Central (brainstem/cerebellum):** Pure vertical, pure torsional, or direction-changing nystagmus; non-fatigable; NOT suppressed by fixation; may have minimal vertigo but prominent neurological signs. Causes: stroke (AICA/PICA territory), MS, tumour, Wernicke's encephalopathy.

## §10 OLFACTION (SMELL)

| Component           | Details                                                                                                                                                                                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Receptor Neurons    | ~10-20 million bipolar olfactory receptor neurons (ORNs) in olfactory epithelium (superior nasal cavity); each expresses one of ~400 human olfactory receptor (OR) genes; ciliated dendrites project into mucus; axons form CN I bundles                                              |
| Olfactory Receptors | GPCRs (Gs-coupled → ↑ cAMP → opens cyclic nucleotide-gated channels → Na <sup>+</sup> + Ca <sup>2+</sup> influx → depolarisation → AP in ORN). Each OR responds to multiple odorants; each odorant activates multiple ORs → combinatorial code gives discrimination of ~10,000 odours |
| Olfactory Bulb      | Glomeruli: ~2,000 spherical structures where ~25,000 ORN axons expressing the same OR converge onto ~1 mitral cell + tufted cells; first processing station; lateral inhibition by periglomerular and granule cells sharpens tuning                                                   |

| Component                                       | Details                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Olfactory Pathway (NO THALAMIC RELAY — unique!) | 1. Primary olfactory cortex (pyriform/piriform cortex + periamygdaloid area + entorhinal cortex) — direct projection from olfactory bulb via olfactory tract; 2. Amygdala (emotional response to smells); 3. Entorhinal cortex → Hippocampus (olfactory memory — very persistent!); 4. Orbitofrontal cortex (conscious smell perception + odour-flavour association); 5. Hypothalamus (feeding behaviour) |
| Olfactory Renewal                               | ORNs are replaced every 30–60 days (sustained neurogenesis from basal cells) — one of few examples of adult neurogenesis in humans                                                                                                                                                                                                                                                                        |

#### ANOSMIA — CLINICAL SIGNIFICANCE

**Anosmia (inability to smell) can indicate:**

**Sinonasal disease (rhinitis, polyps) — most common; usually reversible.**

**Head trauma — shearing of olfactory nerve filaments as they pass through cribriform plate.**

**Early Parkinson's disease and Alzheimer's disease — olfactory bulb changes are among earliest pathological changes (Braak stage 1 for PD).**

**Kallmann syndrome — congenital anosmia + hypogonadotropic hypogonadism (GnRH neurons normally migrate alongside olfactory neurons).**

**COVID-19 — anosmia/parosmia due to support cell damage; usually recoverable.**

## §11 GUSTATION (TASTE)

| Component            | Details                                                                                                                                                                                                                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Taste Buds           | ~10,000 taste buds in adults; on fungiform papillae (anterior 2/3 tongue), circumvallate papillae (posterior 1/3), foliate papillae (lateral tongue edges), palate, epiglottis, pharynx; each bud contains 50–150 taste receptor cells (TRC) + basal stem cells                     |
| Taste Receptor Cells | Three types: Type I (support; salt-sensing? via ENaC?); Type II (sweet, bitter, umami via T1R and T2R GPCRs → PLC → IP3 → Ca <sup>2+</sup> → ATP release); Type III (sour via H <sup>+</sup> ion channels — voltage-gated; also synapse directly with afferent nerve via serotonin) |

### 11.1 Five Primary Tastes & Their Mechanisms

| Taste  | Receptor Mechanism                                                                               | Function                                                          | Thresh old | Notes                                                                                                      |
|--------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------|
| Sweet  | T1R2+T1R3 heterodimer (GPCR class C) → Gα-gustducin → PLC → IP3 → Ca <sup>2+</sup> → ATP release | Signals caloric content; carbohydrates/sugars; amino acids (some) | Low        | Sucrose, glucose, fructose, saccharin, aspartame, stevia all activate T1R2/3                               |
| Bitter | T2R family (~25 genes) (GPCRs) → same Gα-gustducin/PLC/Ca <sup>2+</sup> pathway                  | Aversion signal for potentially toxic alkaloids/toxins            | Very low   | Most sensitive taste (protective); quinine, caffeine, hops; T2Rs expressed in airways too (immune defence) |

| Taste            | Receptor Mechanism                                                                                                                            | Function                                                                 | Thresh old    | Notes                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| Umami (Savour y) | T1R1+T1R3 heterodimer (GPCR) → similar pathway                                                                                                | Signals protein content (L-glutamate, 5-IMP, 5-GMP)                      | Low           | MSG taste; synergism between glutamate and nucleotides (MSG + mushrooms/fish = strongly umami)                                |
| Sour             | Protons (H <sup>+</sup> ) inhibit K <sup>+</sup> channels OR activate OTOP1 proton channels on Type III cells → depolarisation → 5-HT → nerve | Signals acidic/spoiled food; also attractive at low levels (citric acid) | Low           | Mediated by OTOP1 channel (recently identified); independent of GPCR pathway                                                  |
| Salty            | ENaC (epithelial Na <sup>+</sup> channel) on Type I cells → Na <sup>+</sup> entry → depolarisation → afferent firing                          | Signals Na <sup>+</sup> /electrolyte content for homeostasis             | Interme diate | Amiloride blocks ENaC → eliminates salty taste; low Na <sup>+</sup> state → increased salt appetite (aldosterone sensitises?) |

## 11.2 Taste Pathway

| Segment                     | Details                                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anterior 2/3 tongue         | Fungiform papillae → chorda tympani branch of CN VII → geniculate ganglion → solitary nucleus (nucleus tractus solitarius, NTS)                                                       |
| Posterior 1/3 tongue        | Circumvallate/foliate papillae → CN IX (glossopharyngeal) → inferior petrosal ganglion → NTS                                                                                          |
| Palate, pharynx, epiglottis | CN X (vagus) → nodose ganglion → NTS                                                                                                                                                  |
| NTS → Thalamus              | Nucleus solitarius → ipsilateral VPMpc (ventral posteromedial parvocellular nucleus of thalamus) — taste is one of few senses that is NOT crossed to contralateral thalamus initially |
| Thalamus → Cortex           | VPMpc → Primary Gustatory Cortex (insula and frontal operculum) → Secondary Gustatory Cortex (orbitofrontal cortex — flavour perception, food reward)                                 |